



Estimating tree age depending on tree height for Norway spruce, Scots pine, European beech, and sessile and common oak in Central Europe

Hans Pretzsch^{1,2} · Astor Toraño Caicoya¹ · Miren del Río³

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Abstract

The age of trees in forests, open land, or urban areas is important for biodiversity monitoring, sustainable management, or hazard assessment; but it is often unknown. The height of trees, on the other hand, is often known from remote sensing or can be measured easily and non-destructively. However, there is a sigmoidal relationship between tree age and height, whose inverse function can be used to estimate age from height. Here, we compiled measurements of tree height and age from pre-dominant and dominant trees on long-term experimental plots for the parameterization of a density- and site index-dependent height-age function, tree height = f_1 (tree age, site index, stand density). The study focused on Norway spruce (*Picea abies* (L.) Karst.), Scots pine (*Pinus sylvestris* L.), European beech (*Fagus sylvatica* L.), and sessile and common oak (*Quercus robur* L. and *Quercus petraea* (Matt.) Liebl.) in Central Europe. The extended Chapman-Richard function was used and its inverse function tree age = f_2 (tree height, site index, SDI) enables age estimation from tree height. To quantify prediction uncertainty, a non-parametric bootstrap was applied to the inverted model, providing height-dependent 95% confidence intervals that account for both parameter variability and the non-linearity of the inversion. The accuracy of the age estimation was higher for low and medium height trees (11 – 21%) and decreased with tree height (14 – 22%), a pattern confirmed by the bootstrap-derived confidence intervals. We discussed the application possibilities and limitations of the functions for estimating tree age in the context of monitoring, inventory, evaluation or management of forests or urban trees.

Keywords Chapman-Richard function · Inverse function · Age-height relationship · Tree height-stand density relationship · Quantile regression · Non-parametric bootstrap

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✉ Hans Pretzsch
hans.pretzsch@tum.de

Astor Toraño Caicoya
astor.torano-caicoya@tum.de

Miren del Río
delrio@inia.csic.es

¹ Chair for Tree Growth and Wood Physiology, Department of Life Science Systems, TUM School of Life Sciences, Technical University of Munich, Hans-Carl-Von-Carlowitz-Platz 2, 85354 Freising, Germany

² Sustainable Forest Management Research Institute iuFOR, University Valladolid, Valladolid, Spain

³ Instituto de Ciencias Forestales (ICIFOR-INIA), CSIC, Ctra a Coruña Km 7.5, 28040 Madrid, Spain

Introduction

For many purposes of monitoring, inventory, evaluation or management of forests or urban trees, knowledge of tree age is required. However, tree age is often not known or can only be determined with tree ring analysis, destructive sampling or research in historical records. Tree height, on the other hand, is often known, especially when using common remote sensing methods (Heurich et al. 2003; Mette et al. 2004; Straub et al. 2013; Stepper et al. 2015, 2017; Tian et al. 2017).

For example, for the assessment of the biodiversity of forests (Dieler et al. 2017, Liang et al. 2016) or the biotope value of trees in the context of evaluation, conservation, esthetics, and management (Moning and Müller 2009; Chirici et al. 2012; Hämäläinen et al. 2024; Koike et al. 2024), age is important. For epigenetic studies, tree age is necessary to assess the somatic epi-mutation rates and adaptability of

trees to climatic conditions (Magnani et al. 2000; Johannes and Schmitz 2019; Johannes 2025, Zhou et al. 2025). Similarly, many growth models and management tools (e.g. yield tables) are age-dependent, requiring the knowledge of age for their application (Schober 1975). In the case of urban trees, age information is often missing, but this is relevant for assessing their biotope value and hazard potential, for example (Moser et al. 2015; Rötzer et al. 2020; Franceschi et al. 2022). Age data are also needed as a basis for planning sustainable urban greenery and corresponding models (Manfra et al. 2022; Luetge and Buckeridge 2023; Pretzsch et al. 2023a).

For tree age detection Rohner et al. (2013) distinguished direct methods (tree coring and discs by felling trees) and indirect methods that do not require invasive or destructive sampling. Within the indirect methods, estimation of tree age from tree size is the most common (Rozas 2003; Rohner et al. 2013; Zhang et al. 2022). As tree diameter is easy to measure and tree age information is often obtained by tree ring, i.e. linked to tree diameter development, most of the studies aiming to estimate tree age used tree diameter as indicator. However, there are some few studies using other tree variables such as tree height or crown diameter (Kalliovirta and Tokola 2005) also including environmental variables (Lu et al. 2025). Nowadays, tree height is easily measured by remote sensors (e.g. ALS), so some approaches have developed functions to estimate stand age from tree height and further apply them to remote sensing data for mapping stand age at larger scales (Zhang et al. 2014; Yang et al. 2020; Tian et al. 2024). Yang et al. (2020) directly developed a function to estimate stand age from average tree height, whereas Zhang et al. (2014) developed biomass as function of height, and later age from biomass, inverting the respective functions.

Most of the available studies using height as independent variable were developed to estimate stand age. However, for some objectives it can be necessary to estimate tree age. In a given stand, tree height can vary with stand density (e.g. Meredieu et al. 2003; Toraño Caicoya and Pretzsch 2021), so including stand density in models to estimate age from tree height can improve the results. Similarly, tree height growth is strongly influenced by site conditions (e.g. Assmann 1961; Batho and Garcia 2006; Nord-Larsen et al. 2009), so including of site quality may also improve results.

Box 1: Overview of the social tree classes in forest stands (Assmann 1961; Kraft 1884) and that only tree classes 1 and 2 were included in this study.

Technical term	Class	Definition	Handled in this study
Pre-dominant trees	1	Very high trees with exceptionally well-developed crowns	Included
Dominant trees	2	High trees with relatively well-developed crowns	Included
Co-dominant trees	3	Heights lower than class 2, crowns with onset of degeneration	Excluded
Dominated trees	4	Crowns restricted on all sides or on two sides and dying	Excluded
Overtopped trees	5	Entirely overtopped trees; dying or already dead	Excluded

We use the relationship between stand age and stand mean height in Fig. 1a to show the basic pattern and species-specific differences. Figure 1b shows the inverse curves with stand height scaled on the x- and stand age on the y-axis. The curves were derived from common yield tables (e.g. Schober 1975). However, they are not suitable for tree age assessment because they are based on stand-specific values. The stand ages are usually lower than that of the individual tree because, depending on the tree species and silvicultural systems, trees that are already several years old are used for stand establishment. The mean height are the average of trees of all five social tree classes (Box 1) including co-dominant, dominated, and overtopped trees, whose height development has slowed down as consequence of competition. Due to competition-related mortality, the mean and dominant heights increase and thus usually precede the individual height development of pre-dominant or dominant trees (Assmann 1961). Depending on the thinning method used, more or less trees of defined social classes are removed, which shifts the mean height mathematically.

Our question was how the tree age (not the stand age) and the individual tree height (not the middle or top height) are related. The desired functions are intended for estimating tree age as a function of tree height, as can be measured in forests or cities using TLS, for example (Ahmed et al. 2024, 2025). Figure 1 is nevertheless informative, as it shows the basic pattern. In addition, it indicates that the necessary database for deriving tree height-age relationships is available, as yield tables are based on long-term experiments which may provide suitable data. The mean heights shown in Fig. 1 were ultimately derived from height distributions. And from the corresponding stands, there are also the stand-related

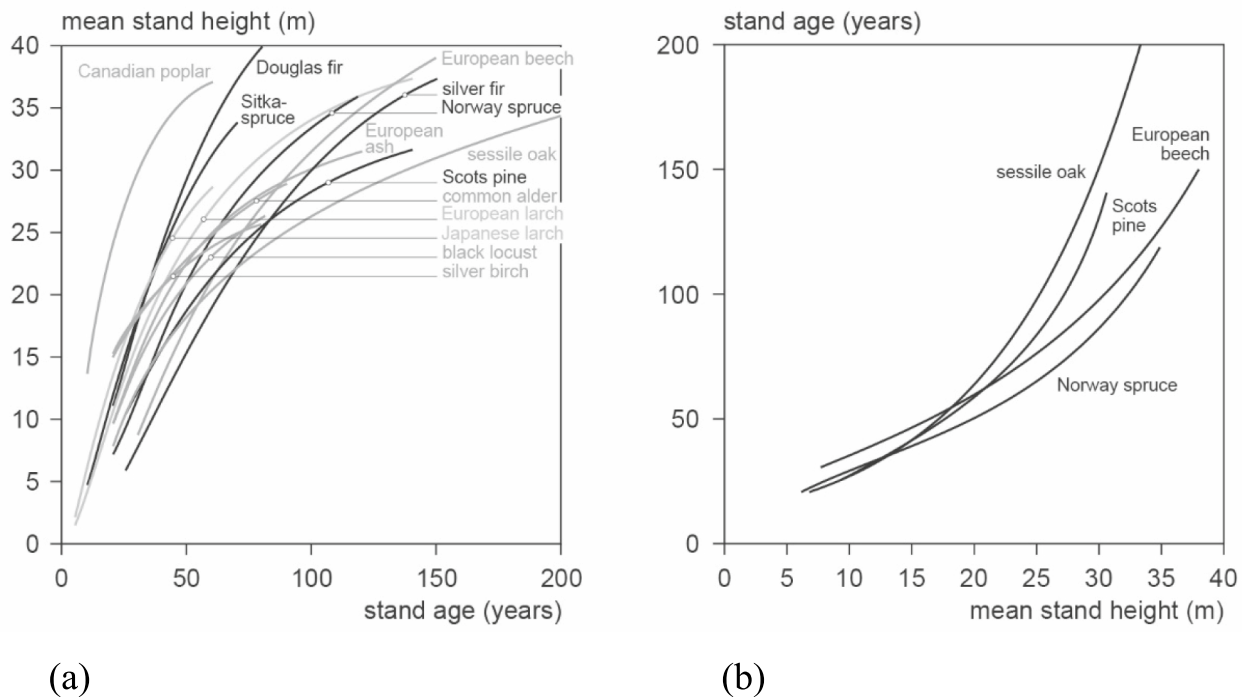


Fig. 1 Basic pattern of the relationship between height and age. **a** Relationship between stand age and mean stand height of fully stocked pure stands on very good sites for various tree species (Schober 1975). **b** Inverse relationship shown for the tree species considered in this study

values such as site index and stand density available. Thus, in the following we used data from long-term experiments, which were also used for the development of yield tables and their age-height curve systems. However, we use the individual data behind the stand values of the tables, the potential of which has hardly been exploited to date.

Based on long term experiments we developed functions for estimating tree age from tree height for Norway spruce (*Picea abies* (L.) Karst.), Scots pine (*Pinus sylvestris* L.), European beech (*Fagus sylvatica* L.), and oak (*Quercus robur* L. and *Quercus petraea* (Matt.) Liebl.). The aim of this study was to parameterize functions with which the age can be estimated as a function of the current tree height. We used long-term measurement series of trees and stands with known development of age and height as a database. The motivation for the present work arose from repeated requests for methods of age estimation based on the current height of trees. We have pursued the following goals in detail:

(i) Measurements of tree height and tree age should be compiled to cover as wide a range of site conditions and stand densities as possible. To eliminate competition-induced growth delays, only pre-dominant and dominant trees were selected from the dataset for further analysis.

(ii) Using this dataset, nonlinear relationships of height development over age, with site quality and stand density as covariates should be parameterized.

(iii) By forming the inverse function, we should obtain functions that relate tree age to tree height, stand density, and site quality. Error calculations should reveal the accuracy of the age estimates for different tree heights.

The possible applications and limitations of the functions in forests, open land and urban areas should be discussed.

Methods

Data base

In order to develop suitable age-height relationships, we needed a broad database of the height and age of trees and the corresponding stand variables. Therefore, long-term experimental plots were selected from which such parameters were known over long periods of time. As tree height development depends on site quality and stand density, we chose experimental plots that covered a wide range of sites and different density levels. Thus we selected long-term experimental plots for the tree species Norway spruce, Scots pine, European beech and sessile and common oak (Table 1).

We pooled common oak (*Quercus robur* L.), and sessile oak (*Quercus petraea* (Matt.) Liebl.). The taxonomic status of these oak species has long been subject to ongoing discussions and repeated reassessment (Aas 1991). Common and

Table 1 Overview of the long-term thinning experiments used in this study

species	experiments	plots	plot size ha	first survey	last survey	tree age last survey	thinning regimes
N. spruce	DEN 5, SAC 67, SAC 68, ZUS 603, DEN 606, SAC 607, FFB 612, WBU 613, VOH 622, TRS 639	77	0.09–0.25	1882–1993	1990–2022	48–147	A-E
S. pine	BAY 51, BAY 52, KUL 53, SNA 57, SNA 58, FLA 70, BOD 610, WIE 611	48	0.10–0.30	1899–1987	2014–2021	53–173	A-E
E. beech	FAB 15, HAI 27, STA 91, ARN 638	15	0.10–0.36	1870–1991	2014–2020	82–198	A-E
oak	WAL 88, ROH 90, ROH 620, GEI 649	24	0.16–0.28	1934–2004	2013–2022	38–151	A, D, E

sessile oak have either been described as two distinct species, *Q. robur* L. and *Q. petraea* (Matt.) Liebl. respectively, or are currently placed within the species *Quercus robur* L. as two subspecies *Q. r. robur* and *Q. r. petraea* (Roloff et al. 2008, pp. 506–507). To avoid possible taxonomic pitfalls, we used “oak” as a generic term summarizing both taxa.

All the experimental plots included are located in central and southern Germany and belong to the network of long-term experimental plots managed by TUM for the Bavarian State Ministry for Nutrition, Agriculture, and Forestry. With measurement series since 1870, this network provides a solid database for the derivation of height-age relationships.

This study leveraged 26 long-term thinning experiments located in southern Germany, involving 164 plots and 290 surveys in even-aged, monospecific stands of Norway spruce, Scots pine, European beech, and oak. The plots, originating from either planting or seeding, are situated across diverse environmental conditions, with elevations ranging from 340 to 840 m above sea level, annual precipitation between 640 and 1200 mm, and mean temperatures of 6 to 8 °C. Soil quality across these sites varies from poor to rich. For more detailed geographical information and site characteristics of the long-term thinning experiments in Norway spruce, Scots pine, European beech, and oak sampled in this study see Supplementary Table S1.

Plot sizes varied, with the smallest being 0.09 ha and the largest extending to 0.36 ha. Each experiment consisted of up to 24 plots. Depending on the timing of the first (ranging between 1870 to 2004) and last (occurring between 1990 and 2022) surveys, each experiment underwent between 4 and 18 surveys. The length of observation lasted up to 150 years, and the stand age differed between 36 and 198 years (Table 1), thereby offering a comprehensive dataset for our analysis.

Supplementary Table S2 shows in more detail the number of plots and surveys, year of the first and last survey, stand age at the last survey, and covered thinning grades of the long-term thinning experiments of Norway spruce, Scots pine, European beech, and oak sampled in this study.

The experiments encompassed plots subjected to various thinning grades as defined by Wiedemann (1935) and elaborated in detail by Kramer (1988, pp. 179–183), implemented in this study’s experimental plots (Table 1). The thinning grades are categorized as A-, B-, and C-grade, representing slight, moderate, and intense thinning from below. On A-grade plots, only dead or dying trees were removed. B- and C-grade plots involved the removal of mainly small trees, with B-grade leaving only pre-dominant, dominant, and co-dominant trees and C-grade retaining only the pre-dominant and dominant trees. D- and E-grade thinning denote moderate and strong thinning from above, removing mainly pre-dominant and dominant trees to promote the growth of the remaining trees. The key distinction between D- and E-grade lies in the horizontal distribution of the interventions (D-grade being uniformly distributed, E-grade concentrated around future crop trees) rather than the extent of density reduction (Wiedemann 1935). We therefore grouped D- and E-grade plots in our evaluation. For a more comprehensive explanation of the internationally defined thinning grades, see Assmann (1970) and Verein Deutscher Forstlicher Versuchsanstalten (1902, 1873).

To determine an integrated measure for site quality, the site index for each plot and survey was determined using yield tables specific to each species: N. spruce by Wiedemann (1936/42), S. pine by Wiedemann (1943), E. beech by Schober (1975), and oak by Jüttner (1955). It is crucial to clarify that the site indexes referenced in this study represent the expected mean stand heights at an age of 100 years.

For quantifying the stand density we used Reineke’s stand density index ($SDI = N \times (25/dq)^a$) with N being the tree number of trees per hectare, dq the quadratic mean stem diameter, and a the species-specific allometric scaling coefficient (Pretzsch and Biber 2005) for Norway spruce ($a = -1.664$), Scots pine ($a = -1.593$), European beech ($a = -1.789$), and oak ($a = -1.424$), respectively. For the modelling we used the SDI values at the stand level calculated for each plot and each survey. We used the SDI because it is a widely applied stand density measure (Sterba 1987) and only the most readily available variables, tree number

and mean diameter, are required for its calculation (Reineke 1933).

Stand density indices of $SDI = 100\text{--}300$ apply to plots with solitarily growing trees, SDI values of about $SDI = 500$ for actively managed forest stands, SDI values of about $SDI = 1000$ represent conservatively managed, dense stands. Values of $SDI = 1000\text{--}2000$ represent extremely dense and tightly packed stands, such as those created for experimental purposes on long-term trial plots.

All included stands are even-aged. The beech stands established via natural regeneration were established in one year, virtually from a single cast. They are therefore like the planted or sown stands even-aged. In the case of the planted stands, the tree and stand ages differed by several years as plants were up to 4 years old. For all stands, both the age of the plants at the time of establishment and the year of establishment were known. This enabled us to base our evaluation on known, documented age data for the trees. The compilation of the available tree and stand data of the long-term experiments resulted in a dataset with altogether $n = 218,553$ individual tree records.

Data preparation

The data preparation started with the elimination of outliers, mainly trees with unrealistic heights larger than 50 m, trees with crown breakage, or trees with statistically implausible ratios between height in m and diameter in cm of $h/d > 5$. Of the remaining trees, not all were included in the evaluation, but only the 20% tallest of the respective survey on each plot. In this way, we eliminated trees with a height development delayed by competition and misleading age-height relationship. A quantile linear regression between tree height and tree age with $\tau = 0.80$ was calculated for this purpose by plot, following Eq. 1 with h , individual tree height; age, tree age; a and b , model coefficients.

$$\ln(h) = a + b \times \ln(\text{age}) \quad (1)$$

Trial calculations using nonlinear quantile regression based on Eq. 2 did not yield satisfactory results for many experiments and plots. The following potential difficulties in applying nonlinear quantile regression, which have also been reported in the literature (Koenker and Park 1996; Furno 2007; Uematsu 2019), were confirmed. The choice of computable initial values proved difficult, different initial values had a major influence on coefficient estimation, the optimization algorithm often did not converge, and in several cases there were multiple local optima instead of a global one, so that parameter estimation was often not possible or yielded implausible values. Therefore, we chose the more stable method of linear quantile regression based on Eq. 1 for sorting out socially disadvantaged trees.

In this way, the original data set was reduced to about 20%, sorting out the socially inferior trees, but including pre-dominant and dominant trees in each plot. As part of some of the previous plot surveys, the social classes of all trees were also determined. Using this data, we were able to ensure that the choice of the $\tau = 0.80$ quantile included the pre-dominant and dominant trees and excluded the co-dominated, dominated, and overtopped trees from the analysis (Table 2). This reduced data set was used to derive the height-age relationship (see Table 3).

Model formulation

Among the many growth functions for trees (Zeide 1993), the Chapman-Richard function (Richards 1959; Shifley and Brand 1984; Zhao-gang and Fengri 2003) is particularly adaptable.

$$h = A \times (1 - e^{(k \times \text{age})})^m \quad (2)$$

The three parameters A , k , and m make it possible to simulate the asymptote, slope and inflection point of size development over age. For example, the frequently used double-logarithmic parabola (Assmann and Franz 1965), the function by Bertalanffy (1951) or the logistic function

Table 2 Database for the development of the tree height-age relationship with site index, SI, and stand density index, SDI, as covariates. SI, site index in terms of mean stand height at age 100. SDI, stand density index according to Reineke (1933) (see Sect. "Data base")

Species	N	Height			Tree age			SI			SDI		
		Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
		m	m	m	yrs	yrs	yrs	m	m	m	ha ⁻¹	ha ⁻¹	ha ⁻¹
N. spruce	26,876	20.6	1.4	53.0	52	11	143	37	27	48	861	76	1613
S. pine	114,239	8.7	0.7	47.8	37	17	148	23	17	34	869	277	3156
E. beech	4538	28.4	2.0	44.2	113	45	188	31	24	36	862	368	1420
oak	72,900	12.1	1.6	43.0	59	21	142	28	22	32	1076	253	2187

Shown are the number of included plots, the range of the plot size, year of the first and last survey, the tree age at the last survey, and the represented thinning regimes A, reference plots; B and C moderate and heavy thinning from below; D and E moderate and heavy thinning from above (see explanations in Sect. "Data base")

Table 3 Characteristics of the model $h = (a \times \ln(SI) + b \times \ln(SDI)) \times (1 - e^{(c \times \text{age})})^d$, parameterized for the tree species Norway spruce, Scots pine, European beech, and oak

Tree species	n	a	se(a)	b	se(b)	c	se(c)	d	se(d)
Norway spruce	5250	24.48***	0.28	- 5.63***	0.14	- 0.02***	0.0003	1.42***	0.01
Scots pine	22,296	23.91***	0.18	- 6.22***	0.06	- 0.02***	0.002	1.17***	0.01
European beech	889	15.81***	0.32	- 1.95***	0.16	- 0.02***	0.0006	1.15***	0.05
oak	9359	15.73***	0.92	- 0.99***	0.11	- 0.009***	0.001	0.99***	0.06

(Fischer and Schönfelder 2017) are less flexible in the simulation of species-specific slopes or inflection points.

To take the influence of site conditions and stand density into account, the Chapman-Richard function was extended as follows: the asymptote A, which stands for the maximum achievable height, was expanded by the term $a \times \ln(SI) + b \times \ln(SDI)$. In this way, we took into account that the maximum tree height can be modified by the site conditions (Assmann 1961; Nord-Larsen et al. 2009) and the stand density (Meredieu et al. 2003; Toraño Caicoya and Pretzsch 2021). The model thus extended to four coefficients a, b, c, and d.

$$h = (a \times \ln(SI) + b \times \ln(SDI)) \times (1 - e^{(c \times \text{age})})^d \quad (3)$$

This model was parameterized separately for all four tree species in order to simulate their species-specific growth trajectories. For example, the size development of light tree species such as pine or oak is faster, with a greater gradient, earlier turning time and lower final height (pioneer tree species). In contrast, the tree height of shade tree species such as spruce or beech increases more slowly, culminates later and the final heights are greater (climax tree species).

In order to include the effect of the site quality and density, the site index, SI, and the stand density index, SDI, were calculated for each tree from the stand data (for calculation of site index, SI, and stand density index, SDI, see Sect. "Data base").

By forming the inverse function, the following function for estimating the age from the tree height was obtained.

$$\text{age} = \ln(1 - (h / (a \times \ln(SI) + b \times \ln(SDI)))^{1/d}) / c \quad (4)$$

By inserting tree height, site index, and stand density index and by using the species-specific model coefficients a-d tree age can be estimated.

We did not parameterize the inverse function (Eq. 4) but the height-age relationship (Eq. 3) because the latter represents the underlying cause-effect relationship with tree age as the independent variable and tree height as the dependent variable. In this way, it was taken into account that the tree age was known and hardly subject to error, while the height is a more error-prone variable. The regression analysis thus minimized the quadratic mean squares of the heights from the model (y-axis residuals), not the quadratic mean squares of the tree ages (x-axis residuals).

In order to calculate the accuracy of the age estimate, the function values and original measured values were compared. For this purpose, the root mean square error was calculated for three strata, tree heights < 15 m, 15–30 m, and > 30 m, because the error naturally increases with increasing age.

$$m_x = \sqrt{\frac{\sum_{i=1}^n (\text{age}_{\text{esti}} - \text{age}_{\text{obsi}})^2}{n-1}} \quad (5)$$

$$m_{x\%} = m_x / \text{age}_{\text{obs}} \times 100 \quad (6)$$

The variables were age_{obs} , observed tree age; age_{esti} , age estimated by Eq. (4). This resulted in the absolute error in terms of years (m_x) and the relative error in percent ($m_{x\%}$). In particular, age estimation on the basis of height at higher ages is made more difficult by, among others, stagnation of height growth, top drought or snow-breakage of the tree top.

The upstream outlier check and the quantile regression eliminated 83% of the data. And these are in particular the measured values of the trees of the social classes 3–5 (see Box 1), whose height is particularly dependent on neighboring trees. The classification of the data according to plots was important, as this allows the different stand densities to be taken into account, while the inclusion of different experiments in different regions provided the different site conditions.

Our data had temporal and spatial dependencies. Temporal dependencies exist because given test areas and trees were recorded repeatedly. Spatial dependencies exist because the trees included in the study were clustered on plots. We reduced the temporal and spatial dependencies between the measurement variables by using only a subset of the data for the analysis. Of the total $n = 218,553$ individual tree records (Table 1), $n = 37,794$, i.e. approximately 17%, were finally used for model parameterization due to outlier elimination and quantile approach. The selected non-linear model cannot be linearized, so we applied the *nls* routine in *stats* (Bates et al. 2015) for parameter estimation.

Due to the non-linear and non-symmetric nature of the height-age relationship and its inversion, analytical derivation of confidence intervals is often impractical or inaccurate (Zeide 1993). To overcome this limitation, we implemented a non-parametric bootstrap procedure to quantify the uncertainty in inverted age estimates (Efron and Tibshirani 1993). This method complements fixed-range uncertainty assessments by providing a more detailed, continuous depiction of prediction variability across the height spectrum (DiCiccio and Efron 1996). For each species-specific model, 1,000 bootstrap replicates were generated by resampling the original dataset with replacement. Model coefficients were re-estimated for each replicate using nonlinear least squares (Huet et al. 2003). The resulting coefficient sets were applied to the inverse age function across a sequence of target heights, using fixed values of site index (SI) and stand density index (SDI). The distribution of predicted ages at each height enabled the calculation of the 2.5% and 97.5% quantiles, forming 95% confidence intervals around the median prediction. Bootstrap resampling is particularly suitable here

because the inversion of nonlinear growth functions often produces asymmetric error distributions that cannot be well approximated by analytical formulas (Davison and Hinkley 1997). In forestry applications, where growth trajectories vary strongly with site and density, such empirically derived confidence intervals provide a robust and interpretable measure of uncertainty (Temesgen et al. 2008). Such bootstrap-based approaches have been widely used to capture both data variability and parameter sensitivity in forest growth and yield modeling (Gregoire and Valentine 2008; Weiskittel et al. 2011), providing a flexible and robust framework for quantifying uncertainty in height-based age prediction.

For practical application of model 4 we in addition developed auxiliary functions for estimating the site index depending on site characteristics such as altitude above sea level (aasl in m), sum of annual precipitation (p in mm), and mean annual temperature (t in °C). The following functions may be used if local information about the site index is not available; in such cases they can provide SI values as input variable for model 4 for age estimation. We developed the functions for.

Norway spruce
 $\ln(\text{SI}) = 4.01 - 0.38 \times \ln(\text{aasl}) + 0.30 \times \ln(p)$,

Scots pine $\ln(\text{SI}) = -1.17 - 0.31 \times \ln(\text{aasl}) + 0.95 \times \ln(p)$

European beech
 $\ln(\text{SI}) = 1.15 + 0.13 \times \ln(p) + 0.70 \times \ln(t)$, and
 oak $\ln(\text{SI}) = 0.37 + 0.20 \times \ln(p) + 0.76 \times \ln(t)$.

The functions were derived by linear mixed models. These functions represent a secondary aspect of this evaluation. Therefore, we have placed their derivation, the statistical parameters, and the visualization of the functions in the Supplementary Material. The statistical model characteristics are presented in Supplementary Table S3, the model curves are visualized in Supplementary Figure S1, and in the Supplement Text S1 we also briefly explain the derivation of these auxiliary relationships.

For all calculations, we used the statistical software R 4.1.0 (R Core Team 2024), explicitly employing the packages *quantreg* (Koenker et al. 2018) for the quantile regression and *stats* for calculating the non-linear regression analyses (Bates et al. 2015). Normality of errors was verified by making normal q-q plots of the residuals. Residuals were plotted against the fitted values for Norway spruce, Scots pine, European beech, and oak, respectively. They showed that there was no violation of variance homogeneity (see Supplementary Figure S2, a-h). The bootstrap-based confidence intervals for the inverted age predictions were derived using custom scripts based on resampling and repeated model fitting with *nls* (Bates and Watts 1988).

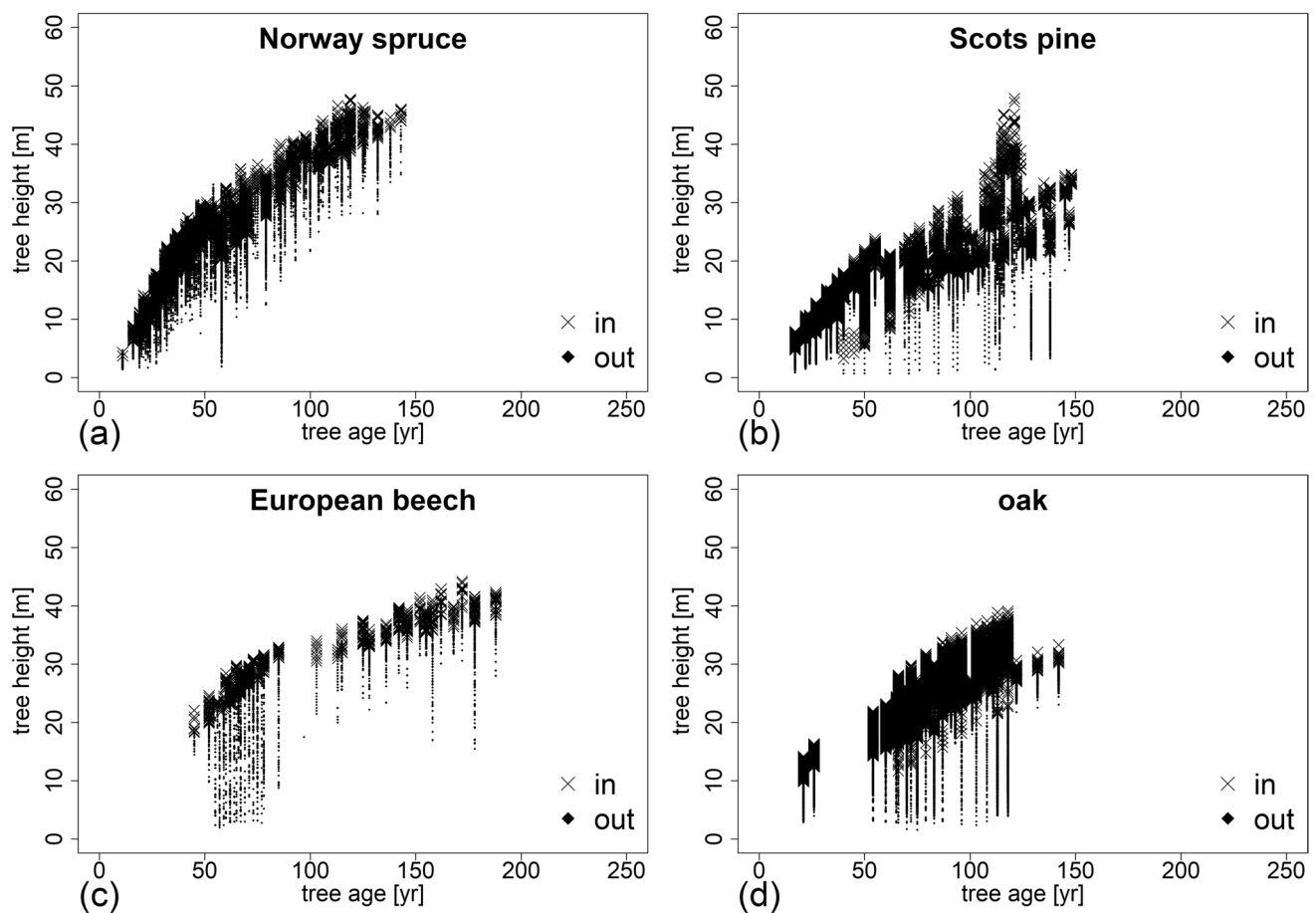


Fig. 2 Visualization of the database for the parameterization of the height-age relationship for **a–d** Norway spruce, Scots pine, European beech, and oak. Only the 80th quantile (the top 20% of trees in each image) was used for parameterization (crosses, in). The small black

diamonds (out) represent co-dominant, dominated, and overtopped trees (classes 3–5 according to Box 1) that were not included in the parameterization

Results

Descriptive results

Figure 2 shows the available measured values in total with highlighting (crosses) of the values that were ultimately used for the parameterization of the height-age relationship. The upper edge lines are very species-specific, for example they are steep in spruce with higher final values (Fig. 2a), in beech the course is flatter, more elongated with lower final heights. The lower edge lines are significantly higher in the shade tree species spruce and beech than in pine and oak. By using only the 80th percentile for further parameterization we only included the trees in the upper 20% height stratum, which Assmann (1961) defined as pre-dominant and dominant from the rest of the stand.

The compilation of the available tree and stand data of the long-term experiments resulted in a dataset with altogether $n = 218,553$ individual tree records summarized in Table 2. For future application of the derived functions it is interesting to know that the mean site index was 37 m for Norway spruce, 23 m for Scots pine, 31 m for European beech and 28 m for oak. The mean SDI values amounted to 861–869 trees per hectare for Norway spruce, Scots pine and European beech, and 1076 trees per hectare for oak.

Results of the parameterization of the extended Chapman-Richard functions

Table 3 shows the results of the model fitting for Norway spruce, Scots pine, European beech, and oak and Fig. 3

visualizes the course of the functions and their inverse function (see Sect. " **Model formulation**", Eqs. 3 and 4).

All parameters of the extended Chapman-Richard function (Table 3) showed clear species-specific differences. The coefficient a in the model is linked to asymptote and shows the effect of site quality on the final height; it was particularly low for oak and highest for spruce. Coefficient b quantifies the effect of stand density on the final height; the reduction was lowest in oak stands and highest in pine stands. The slope coefficient c indicated a significantly flatter development only in oak. In contrast, the position of the turning point, represented by coefficient d , ranged from particularly low values in oak to high values in spruce.

Figure 3, left, visualizes the species-specific trends and shows the high flexibility of the selected function. The inverse curves shown on the right-hand side of the graph represent the estimated result of the evaluation. They are suitable for estimating the tree age as a function of the tree height.

Figure 4 shows how site index values of $SI = 20$ and $SI = 40$ (above and below) modulate tree age. For all species, the tree age (under the otherwise constant conditions) decreased with increasing site index. This is of course due to the fact that all tree species reach a certain tree height faster on better sites, i.e. at a younger age, than on unfavorable sites. The decrease was least pronounced for oak and most pronounced for pine.

On unfavorable sites, the age-height curves of the tree species are closer together than on favorable sites (Fig. 4, b and d). In both cases, however, the spruce, for example, reached a height of 30 m 20–40 years earlier than the other species. Oak and beech were significantly behind in height development.

In the main text, we have shown the relationships between height and age for SDI values of $SDI = 500$, because these are the usual density levels in forestry practice, i.e., in normally managed stands. In Supplement Figures S3 and S4, we have also shown the relationships between height and age for SDI values of $SDI = 1000$ for further information. Such densities are achieved in only moderately thinned or untreated forest stands.

Figure 5 shows clear species-specific differences for both the effect of the site index and the stand density. The effect of the site index was generally more pronounced (Fig. 5a) than the effect of stand density (Fig. 5b). However, the inclusion of both site index and stand density allowed age estimates for a wide range of site qualities and stand densities.

For instance, for very dense stands, SDI values of 1000 trees per hectare can be used and they then provide higher age values, all other things being equal, than stands with densities of 500 trees per hectare or for free-standing trees where the SDI values approach zero.

If all other variables are held constant, stand density had a positive effect on age (Fig. 5b), which was particularly strong for pine, but less pronounced for oak. This means that increasing density retards height growth, i.e. a given height was reached later than in more loosely stocked stands. Stand densities of $SDI = 1000$ occur in only moderately thinned or untreated forest stands. On extremely dense experimental plots the stand densities could become even higher. Stand densities in the medium density range ($SDI = 500$) can be found, for example, in more heavily thinned stands. Low SDI values can be used to estimate the heights of trees in very loosely stocked stands, of solitary or urban trees, which then, *ceteris paribus*, tend to yield lower ages for a given height.

In practical forestry, for the age estimation, the site index can be taken from local documents. Forest management plans, forest inventories, or site maps often reveal the locally or regionally predominant species-specific site index levels without the tree ages in the current stands being known (Seynave et al. 2005; Brandl et al. 2018). By setting high or low densities, the tree ages in stands of different densities or in solitary positions can be estimated.

Accuracy of the tree age estimation depending on tree height and stand co-variables site index, SI, and stand density index, SDI

Table 4 shows that the accuracy of the age estimation was higher for low and medium height trees (11 – 21%) and the accuracy of the age estimation decreased with tree height (14 – 22%).

The fixed-range errors in Table 4 can be complemented using the bootstrap method by providing a continuous view of estimation uncertainty over height in the inverted age estimation across tree heights for the four representative species (Fig. 6). The 95% confidence bands are shown for two site indices ($SI = 20$ and $SI = 40$) under a fixed stand density index ($SDI = 500$). As expected, uncertainty increases with tree height, especially beyond 25–30 m, where the inversion becomes increasingly sensitive to model parameter variation. The deviation from the median age remained close to zero for small trees, confirming the model's stability in early developmental stages. The extent of uncertainty was species-specific, with narrower bands for Norway spruce and European beech and broader bands for oak and Scots pine, particularly under lower site conditions. While the uncertainty bands were wider under $SI = 20$ m, all species showed convergence around the median at lower heights.

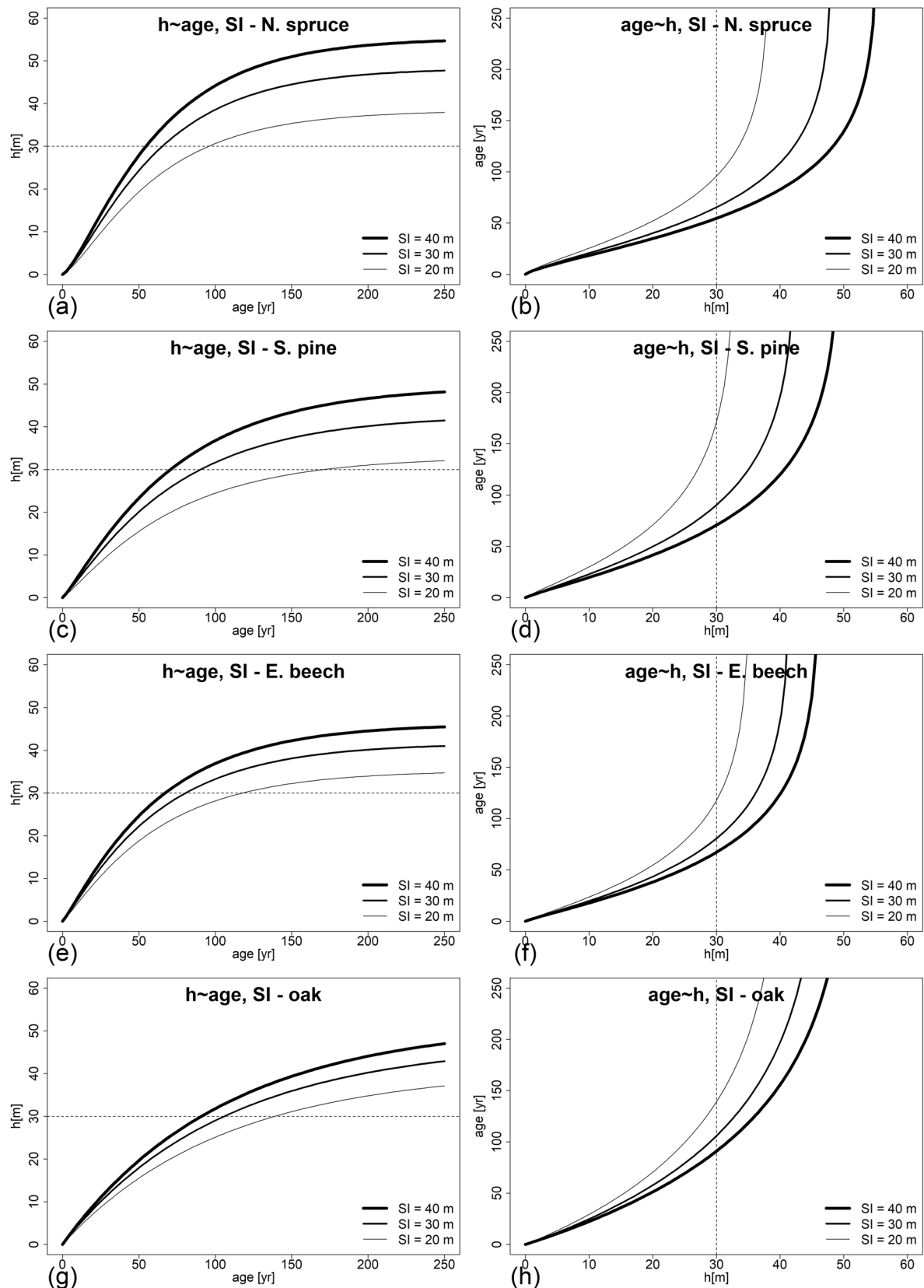


Fig. 3 Visualization of the functions with tree height depending on age (left) and the inverse function with tree age depending on tree height (right). The relationships were shown for **a** and **b** Norway spruce, **c** and **d** Scots pine, **e** and **f** European beech, and **g** and **h** oak. The relationships were shown for different site conditions and a stand density index of SDI=500. The graphs were based on the Eqs. 3 and 4 (see Sect. "Model formulation" and Table 3)

Discussion

Tree age as a function of tree height

We used tree height to estimate age and give it preference over stem diameter because tree height has become easy to measure by new remote sensing airborne technologies such as ALS (Wang et al. 2019). In contrast, stem diameter is hardly directly accessible by airborne technologies (Heinzel and Huber 2017). It requires more complex and costly terrestrial remote sensing measurements (Liu et al. 2018). From airborne measurements stem diameters can only be derived indirectly, by using allometric relationship between tree height and diameter or assuming stem shape models (Pitkänen et al. 2019). We therefore based our estimation on the directly measurable tree height.

Functions are presented with which ages of pre-dominant and dominant trees can be estimated from the current tree height for the most important tree species in Central Europe. They are based on the measurements on long-term experimental plots and represent very accurate and up-to-date age and height information derived for the parameterization of the extended non-linear Chapman-Richard $h = f$ (age, site index, stand density). The estimation of age as a function of height is made possible by forming the inverse functions of these relationships. While our data come from even-aged stands, our analysis relates tree age to the height of individual pre-dominant and dominant trees rather than to stand averages. This individual-tree focus differentiates our approach from previous studies that relied solely on mean stand age and mean height.

The relationships have a number of advantages over the use of yield tables. Yield tables mostly represent historical growth conditions and treatment variants and their validity has decreased (Kimmins 1990; Spiecker et al. 2012; Pretzsch et al. 2023b). Yield tables are based on stand averages and are therefore strongly overlaid by the decrease in tree numbers with increasing age due to self-thinning or active thinning. The mean values shift mathematically as a result, simply because of the collective changes. Finally, yield tables only provide the heights as a function of age, but not the inverse of this relationship, so they do not offer the possibility of determining age directly on the basis of height and any other covariates. Therefore, the functions derived here represent a step forward for applications in the field of monitoring and remote sensing based inventory.

When comparing the developed approach with methods used for dominant height growth curves developed to estimate stand site index (Burkhart and Tomé 2012), the fitted models present some advantages and disadvantages. The main advantage is that they are based on a broader data set that consider different stand densities, as recent studies demonstrate the influence of stand density on tree height (Meredieu et al. 2003; Toraño Caicoya and Pretzsch 2021). A disadvantage is that they assume a similar growth pattern for a given density and site index, thus generating anamorphic height growth curves. The functions represent a broad range of site quality, from unfavorable to very good site conditions. In Central Europe, height ratings of SI=20 m represent the lower and SI=40 m the upper range. For medium growth conditions, a site rating of SI=30 m can be used. Furthermore, the functions represent a very wide range of stand densities. Experiments were included that are very sparsely stocked (solitary conditions), moderately or heavily thinned (B-, C, D-, E-grades) as well as completely untreated and very densely stocked areas (so-called A-grade or 0-plots, which serve as reference in thinning experiments).

When applying the functions SDI values can be taken from forest inventories, forest management documents, or applied silvicultural treatment programs without explicit density measurements (Sterba 1987; Zeide 2005; Pretzsch 2020). To give some rough figures, SDI values of about SDI=500 can be used for actively managed forest stands (see Sect. "Data base"). SDI values of about SDI=1000 can be used for conservatively managed, dense stands and values of SDI=1000–2000 for extremely dense and tightly packed stands, such as those created for experimental purposes on long-term trial plots. SDI values of SDI=100–300 represent solitary conditions on the experimental plots; therefore, such values may also be used to estimate the height-age relationship of open or solitarily growing trees. But of course, any extrapolation beyond the range of values covered by the experimental plot data is subject to error.

The functions are particularly suitable for heights of up to 30 m. As the tree height increases, the accuracy of the age estimation decreases. Particularly in older stands, with a slowdown in height development, crown stumps, top breaks, etc., the indicator value of height for age determination decreases.

The data used for model calibration cover a very wide range of tree heights (0.7–53.0 m), tree ages (11–188 years), site qualities (SI 17–48 m) and stand densities (SDI 76–3156 tree per hectare) (Table 2). This is because these are experiments in which the effects of extreme variants that are unusual in practice are also tested. SDI values may be less precise in operational contexts. However, in forest inventories and management plans, SDI or its components are commonly available. Where uncertainty is larger, this will naturally increase the error in age estimation. Our bootstrap

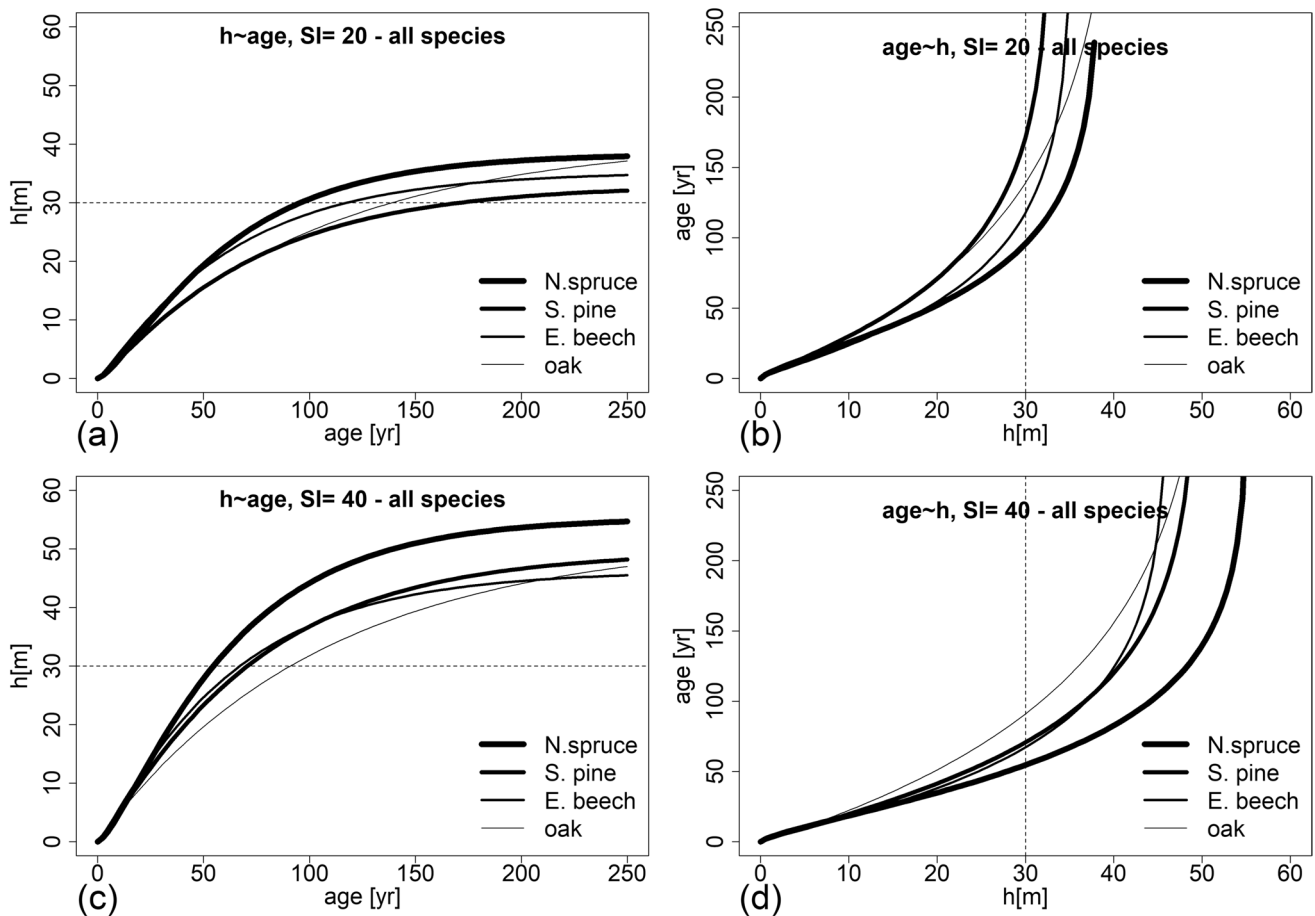


Fig. 4 Height development over age (left) and age over tree height (right) for the four tree species on **a** and **b** unfavorable sites with a height rating of SI=20 m and **c** and **d** on very good sites with a

height rating of SI=40 m. The stand density was kept constant at SDI=500 trees per hectare. The graphs were based on the models 3 and 4 (see Table 3)

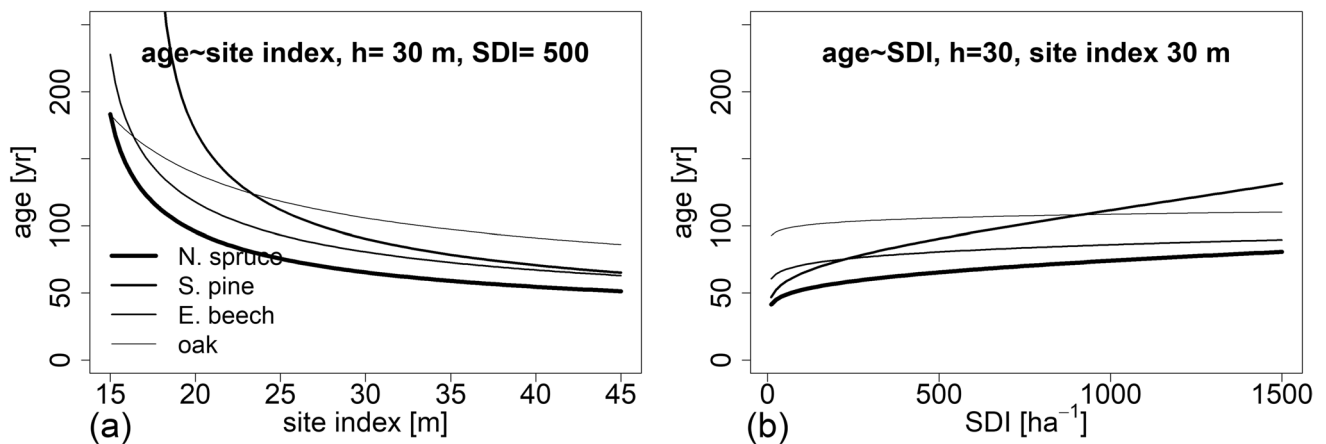


Fig. 5 Effect of **a** site index and **b** stand density on the tree age for a fixed tree height of 30 m shown for Norway spruce, Scots pine, European beech, and oak. The graphs were based on the models 3 and 4 (see Table 3)

Table 4 Overview of the species-specific fixed-range accuracy of age estimation for the three tree height ranges $h < 15$ m, $h = 15\text{--}30$ m, $h > 30$ m. The accuracy is given in percentage with the reference

Species	Height stratum < 15 m			Height stratum $15\text{--}30$ m			Height stratum > 30 m		
	Error abs	Obs mean	Error rel	Error abs	Obs mean	Error rel	Error abs	Obs mean	Error rel
	years	years	%	years	years	%	years	years	%
N. spruce	3	23	11	9	44	21	17	97	18
S. pine	4	26	15	12	71	17	20	126	16
E. beech	7	48	15	12	64	19	19	138	14
oak	5	24	21	13	73	18	23	105	22

error abs absolute accuracy in years, *obs mean* observed age in years, *error rel* % relative error of age estimation for the given tree height strata

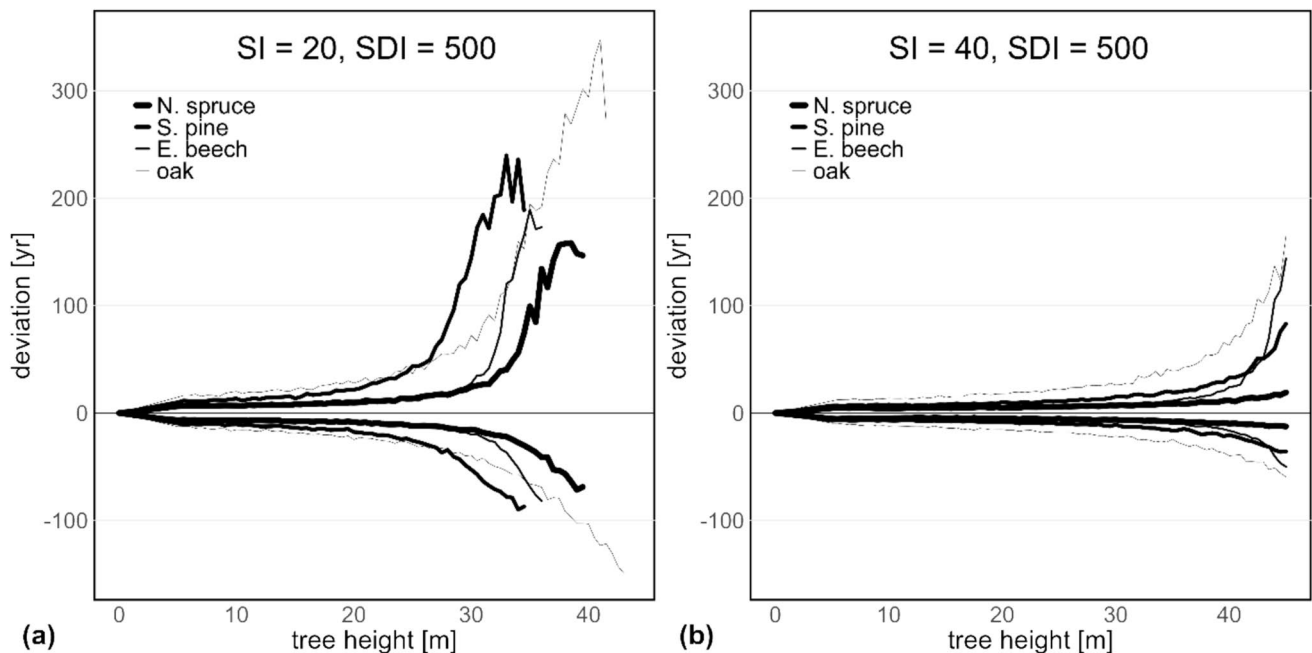


Fig. 6 Normalized uncertainty in inverted age estimation with a bootstrapped 95% confidence intervals four representative species (Norway spruce, Scots pine, European beech, and oak), under fixed stand density (SDI=500) and two site index scenarios: **a** SI=20 m and **b**

SI=40 m. Lines indicate deviation from the median predicted age, with line widths varying by species. Confidence bands reflect uncertainty in both the non-linear model fit and the inversion process, as estimated through resampling

confidence intervals explicitly illustrate how such uncertainty propagates into predictions (Fig. 6).

Of course, the functions only provide rough estimates of age because the growth curve of height over age and the corresponding inverse relationship can vary greatly from location to location. Behind a given site index rating there may be trajectories with different slopes and turning points, resulting in polymorphic height growth curves. For precision, region-specific relationships would therefore be conceivable, which would allow the covariates site index and SDI to also affect the curve coefficients slope (c) and inflection point (d) (see Eq. 3 and 4), as demonstrated for regional (Calama et al. 2003) and climate and soil-dependent

dominant height growth models (Bravo-Oviedo et al. 2008). In the approach used here, these coefficients are species-specific, but within the species, site index and stand density only affect the asymptote, i.e. the height level of the curve (see Eq. 3).

Possibilities and limits of the derived tree age-tree height relationships

Trees reach their maximum height for a given age with medium competition, i.e. with full stocking or moderate thinning. Excessive competition, which occurs particularly in the intermediate and understorey of more structured

stands, can lead to growth retardation. Height stagnation in the understorey over 50–150 years is not uncommon, especially with shade-tolerant tree species. Very high density can also trigger stagnation in dominant height, especially in plantations where homogeneity in spacing limit social differentiation (Montero et al. 1999). In such cases, height loses its indicator role for age. A slowdown in height development can also occur with very little competition, e.g. with permanently solitary mature trees in the field or in urban areas, with heavily exposed trees after high thinning or clearing, or with trees that have survived a wind-throw in their surroundings (Gessner 1961). To account for the effect of low or high competition on height development, we have included stand density in the functions. The age of solitary or urban trees is most likely to correspond to the age-height relation for very low SDI values (Hasenauer 1997).

The functions were developed for fully stocked or moderately thinned stands. With heavier thinning, especially when the canopy is opened after thinning from above, height growth may decrease in favor of stem diameter growth (Niklas 1995; Toraño Caicoya and Pretzsch 2021; Pretzsch et al. 2022). If the age progresses but the height stagnates, the measured height is based on a higher age, i.e. the tree age is systematically underestimated. The functions are not suitable for age estimation in multi-layered stands because they do not take delays or waiting times into account. However, since many of today's understorey-structured stands are dense and relatively dense over long periods of time, the functions should be suitable for the pre-dominant and dominant trees (classes 1 and 2 according to Box 1) in such forests.

Accuracy of tree age estimation based on tree height

Even with the same age, stand density and site quality, the tree height can vary due to factors such as the genetic origin of the trees (Hamrick et al. 1981), the type of stand establishment (artificial versus natural) (Assmann 1961), past abiotic (late frost, top blight) or biotic (insect damage, mistletoe infestation) damage (Jutras et al. 2010; Schmid et al. 2021). In addition, the measurement of tree height, for example, is subject to greater measurement errors compared to diameter measurement. Classic terrestrial measurements based on the geometric or trigonometric principle are subject to measurement errors of 2–6% (Kramer and Akça 1995; Prodan 1965). In our measurement series of up to 150 years, the errors from earlier to current measurements have probably decreased due to the increasing quality of the equipment. Height measurements via remote sensing are also becoming increasingly accurate; the errors in height measurements using ALS are now only 1–2% (Sibona et al. 2016; Ganz et al. 2019; Wang

et al. 2019). In large-scale applications, radar-based satellite systems have enabled regional height estimations across boreal and temperate forests (Abdullahi et al. 2016; Toraño Caicoya et al. 2016), while newer SAR- and LiDAR-based methods provide higher precision in capturing vertical structure, particularly in dense and heterogeneous stands (Toraño Caicoya et al. 2015), offering multi-layer height information that supports a more accurate assessment of stand age distributions.

The determined accuracies of the age estimation from heights are therefore due to the variation in height, even if age, stand density and site conditions are kept constant. On the other hand, they are due to inaccuracies in the height measurement. Seen in this light, age estimation accuracies of 10–20% are understandable and acceptable.

In addition, the bootstrap-based inversion accounts for both parameter uncertainty and the inherent non-linearity of the inversion, resulting in asymmetric and height-dependent confidence intervals. As shown in Fig. 6, the uncertainty is considerably lower on high-quality sites ($SI = 40$) compared to low-quality sites ($SI = 20$), indicating more stable and predictable growth conditions. Moreover, species with larger sample sizes and better representation in the dataset, such as Norway spruce and European beech, exhibit narrower confidence intervals and more robust inversion behavior, particularly in early and mid-developmental stages. This highlights the importance of data coverage for model calibration and confidence assessment (Temesgen et al. 2008). Incorporating these uncertainty estimates into growth assessments and stand reconstructions enhances both ecological interpretation and silvicultural decision-making by providing a more realistic representation of uncertainty in modeling and simulation outcomes.

Since site index (SI) and stand density index (SDI) are included as covariates in the extended Chapman-Richards function (see Eq. 3 and 4); their uncertainty also propagates into the inverted age estimates. Using implicit differentiation of Eq. 4, we derived an analytical expression for the first-order sensitivity $\partial(\text{age})/\partial(SI)$. For illustration, using the parameters fitted for Norway spruce at $h = 25$ m, $SI = 30$, and $SDI = 500$, the model predicts an age of about 52 years. The corresponding local sensitivity is $\delta(\text{age})/\delta(SI) = -1.05$ years/m and the elasticity is -0.61 , meaning that a 1 percent increase in SI decreases the estimated age by roughly 0.6 percent. This shows that moderate SI uncertainty causes only limited variation in predicted age, except at large heights where growth approaches its asymptote.

A similar first-order assessment shows that a $\pm 10\%$ change in SDI affects the estimated age by less than $\pm 5\%$. Because SDI requires diameter information that is not directly available from airborne LiDAR, future remote-sensing applications could use relative spacing index

($RSI = h / \sqrt{\partial N}$) or crown-based competition metrics as practical alternatives.

The present work is a first step towards age estimation; an improvement can be expected by including further covariates in the estimation (crown characteristics, soil variables, climatic conditions).

Perspectives

Similar functions should be parameterized for the age determination of other tree species. However, with the light-demanding tree species (Scots pine and oak) and shade-tolerant tree species (Norway spruce and European beech) selected here, species with fundamentally different age-height relationships are already covered. In the juvenile phase, light-demanding tree species have lower ages behind given heights than shade-tolerant tree species, because light-demanding tree species are then faster-growing. Later, the relationships are reversed. In the advanced development phase, shade-tolerant tree species continue to grow in height for longer and reach higher final heights. While the light-demanding tree species stagnate earlier in height development and the height then loses its indicator value for age, the height of shade-tolerant tree species remains an indicator value for age for longer.

If the tree ages of other species are to be roughly estimated from the height, the functions presented here can also be used as an aid by analogy. For a species that has not yet been parameterized, it is then necessary to decide which growth, i.e. which ontogenesis, it most closely resembles. Its function can then be used. Alternatively, the functions of the two most similar species can be used to estimate the age, which then at least provides an age range.

Future work, building on an even broader set of dendrometric and environmental data, could also incorporate environmental variables directly into estimators, thus replacing the here developed auxiliary functions with an integrated approach.

Conclusions

A first useful approach for the estimation of tree age from tree height was developed. For its parameterization, we used data from long-term growth and yield trial plots, which have not yet been exploited for such questions. Similar long-term experimental plots are available at other institutions for other tree species. The approach can be used to parameterize estimation functions for other tree species.

Future approaches can also include other tree characteristics in the age estimation. The inclusion of characteristics of

the upper tree crown, which are increasingly available from laser scanning, would certainly be useful. Recent studies by Ahmed et al. (2024, 2025) and Bohnhorst et al. (2025) have shown that the shape of the crown contains important information about the past growth of trees and also about their vitality and future growth potential. An upper crown with a conical shape indicates further growth, a spherical upper crown indicates stagnation of height growth. Since the accuracy of our estimation function decreases with increasing age, information on the crown shape could provide an improvement here. This would help avoiding an underestimation of the age estimate from flattening trajectories in cases of e.g. decreasing height growth, top breakage or top drought.

Author Contribution statement HP: conceptualization, funding acquisition, investigation, methodology, formal analyses, original draft, writing—review & editing, visualization. ATC: investigation, methodology, writing—review & editing. MR: methodology, writing—review & editing.

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Data availability The materials described in the manuscript including all relevant raw data will be freely available upon request from the corresponding author.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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